

NETWORK EFFECTS IN DIGITAL ECONOMY

MAIN TOPICS

- Types of network effects
 - Direct vs indirect network effects
 - Cross-side effects in multi-sided platforms
 - Real-world examples (Nepal + global)
 - Strength, limits, and failure modes of network effects
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- Network effects are like a snowball: the bigger it gets, the faster it grows, but if it hits a wall (bad experience, low quality), it can stop or even roll back.
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NETWORK EFFECTS IN DIGITAL ECONOMY

- Why digital platforms grow fast (or fail fast)

Key Points

- Platforms become more valuable as more people use them
 - Drives exponential growth
 - Central to all digital giants (e-commerce, payments, ride-sharing)
 - **“Why don’t we use 20 different messaging apps? Why does one dominate?”**
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DEFINITION OF NETWORK EFFECTS

- *“When the value of a product or platform increases as more users join the network.”*
 - This is unique to digital markets because digital goods scale rapidly at low cost.
 - **Example**
 - WhatsApp: more of your friends → more value
 - Pathao: more drivers → shorter wait times
 - **Idea:**
One user = little value → Many users = huge value.
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WHY NETWORK EFFECTS MATTER

- Accelerates user adoption
 - Creates competitive advantage
 - Produces winner-take-most markets
 - Creates lock-in and switching barriers
 - Makes new startups hard to compete with established giants
 - **Examples**
 - FonePay (dominant QR payment network)
 - Ride-sharing (Pathao/InDrive)
 - Messaging apps (WhatsApp/Viber)
 - Marketplaces (Daraz, Amazon)
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TYPES OF NETWORK EFFECTS

- Direct network effects
 - Indirect (cross-side) network effects
 - Data network effects
 - Technology/standard network effects
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DIRECT NETWORK EFFECTS (ONE-SIDED PLATFORM)

- Value increases when *same type of users* join.

Examples

- **Messaging Apps (WhatsApp, Viber)**
More users = more people to talk to
- **Social Networks (Facebook, Instagram)**
More creators = more content
- **Gaming**
More players = better matchmaking

Why this is powerful:

You *cannot* use the product alone → needs others.

INDIRECT NETWORK EFFECTS (CROSS-SIDE)

- Value grows when *different groups* join a platform.
 - Typical in 2-sided or multi-sided platforms.
 - **Examples**
 - Uber: More drivers → shorter wait times → more riders → more drivers
 - E-commerce: Sellers ↔ Buyers
 - App store: Developers ↔ Users
 - **More sellers → more choices → attracts buyers → attracts more sellers → loop**
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CROSS-SIDE NETWORK EFFECTS DIAGRAM

- Simple idea:
 - More Users → Attract More Developers
 - More Developers → More Apps
 - More Apps → More Users
 - **Platforms to mention**
 - Android ecosystem
 - Payment networks (Visa: merchants ↔ cardholders)
 - Food delivery apps (Foodmandu: restaurants ↔ customers)
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CROSS-SIDE NETWORK EFFECTS

Dynamic Cycle of Drivers, Riders, and Earnings



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DATA NETWORK EFFECTS

- Product improves as more data = more accuracy.
 - Not user-to-user, but user → data → model → better product → more users.
 - More users generate more data → better algorithms → product gets better → attracts more users.
 - **Examples**
 - Google Maps: More people using = better traffic prediction
 - TikTok algorithm improves with more watch data
 - Health tracking apps (Fitbit): better activity models
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Platform Type	Example	How Data Helps
Search engine	Google Search	More searches → better results & suggestions
Maps/Navigation	Google Maps, Pathao	More users → better traffic, routing, ETA predictions
E-commerce	Daraz, Amazon	More users → better recommendations & inventory insights
Social media	Facebook, TikTok	More user activity → better content recommendations
Payments	FonePay, eSewa	More transactions → better fraud detection, faster payments

TECHNOLOGY NETWORK EFFECTS

- When a technical standard becomes more valuable as it becomes dominant.
 - **Examples**
 - USB-C adoption
 - QR-based payment systems
 - Global GSM mobile standard
 - File formats (PDF, MP3)
 - **Use Nepal example:** FonePay dominance means merchants adopt because others already do.
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Platform Type	Example	Why Standard Matters
Messaging apps	WhatsApp, Viber	Everyone uses the same protocol → easy to communicate
Ride-sharing	Pathao, Uber	Same app & platform → easy for riders & drivers to connect
Payment QR standards	FonePay	Merchants & users use same QR → widely accepted
Video streaming	YouTube, Netflix	Same video format/streaming tech → compatible across devices
Social media	Instagram, Twitter	Common post/image/video formats → easy sharing & interaction

PLATFORM GROWTH STAGES

- *Cold Start Problem* — too few users
 - *Critical Mass* — enough users for network effects to kick in
 - *Growth Acceleration* — positive feedback loops
 - *Maturity* — winner-takes-most phase
 - **Examples**
 - TikTok: overcame cold start with content seeding
 - Ride-sharing provides incentives to attract initial drivers
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COLD START PROBLEM

- Platforms are useless in the beginning because there are no users.
 - **Examples**
 - New messaging app: nobody to chat with
 - New ride app: no drivers → no riders → no drivers
 - New marketplace: no sellers → no buyers → no sellers
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HOW PLATFORMS SOLVE COLD START

- **Subsidizing one side** (Uber paying drivers initially)
 - **Exclusive early content** (TikTok seeding videos)
 - **Invite-only strategy** to create hype (Clubhouse)
 - **Using offline networks first** (FonePay onboarding merchants manually)
 - **Nepal example**
Pathao gave bonuses to drivers for first months → increased supply.
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POSITIVE FEEDBACK LOOPS

- “Success breeds more success.”
 - Users → Value → More Users → More Value
 - **Examples**
 - Facebook college launch strategy
 - Daraz: more sellers reduced prices → attracted more buyers
 - **Instagram**: more creators → more content → more users → more creators
 - **Airbnb**: more hosts → more destinations → more guests → more hosts
 - **Uber**: more drivers → shorter wait → more riders → higher driver earnings
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NEGATIVE NETWORK EFFECTS

- When more users reduces value.
- The opposite of what platforms want.

Types

- Congestion (too many users → delays)
- Spam & low-quality content
- Toxicity in communities
- Ride-sharing surge pricing anger

Examples

- Zoom congestion during peak hours
 - Overcrowded marketplaces lowering trust
 - Too many ads on Facebook reducing engagement
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STRENGTH OF NETWORK EFFECTS

- Strength depends on:
 - How much value each new user adds
 - Whether users are interchangeable
 - Whether switching is easy
 - Platform differentiation
 - **Compare**
 - Strong: WhatsApp (everyone uses it)
 - Weak: new social media clones
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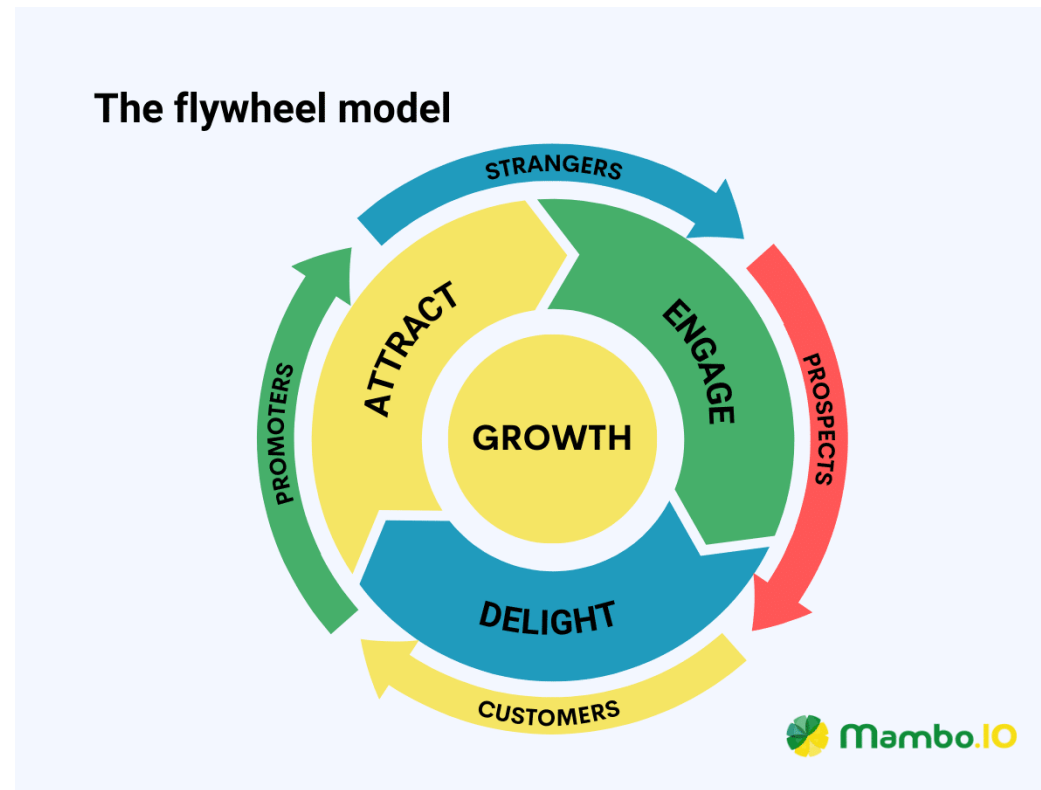
CASE STUDY: RIDE-SHARING IN NEPAL

- **Pathao, InDrive, Tootle**

Why Pathao wins:

- Strong cross-side effects
 - Bigger fleet → faster pickup
 - Larger demand → drivers earn more
 - Spillover: ecosystem for food delivery, courier
 - **The "flywheel model."**
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FLYWHEEL MODEL



CASE STUDY: TIKTOK

- Data-driven recommendations
- Creator → viewer → creator loop
- Viral loops (duets, trends)
- Low effort content creation
- **Result:**
Fastest global adoption curve in history.

CASE STUDY: FONEPAY (NEPAL)

- Merchant ↔ customer cross-side effect
 - Banks integrated → faster adoption
 - Standard QR → strong technology network effect
 - Low transaction cost → accelerated usage
 - **Outcome**
Dominated Nepal QR market early.
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CASE STUDY: GLOBAL EXAMPLES

- **Amazon:** more sellers → more products → more buyers
 - **Airbnb:** more hosts → more destinations → more guests
 - **YouTube:** creators ↔ viewers ↔ advertisers
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WEAK OR BROKEN NETWORK EFFECTS

- Users can easily switch (multi-homing)
 - Low switching cost
 - Poor product experience
 - Better competitor ecosystem
 - **Examples**
 - Google+ failed even with large Google user base
 - Clubhouse lost to Instagram Reels & Twitter Spaces
 - Poorly designed marketplaces die quickly
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NETWORK EFFECTS VS VIRAL EFFECTS

- **Network Effects**

- Product becomes more useful as more people use it.
- Long-term advantage.

- **Viral Effects**

- Product spreads through sharing but not necessarily more valuable.

- **Example**

- Meme goes viral but doesn't improve platform value.
 - WhatsApp becomes more valuable as more users join (network effect).
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NETWORK EFFECTS & COMPETITIVE ADVANTAGE

- **Content**
 - Creates **barriers to entry**
 - Creates **user lock-in**
 - Leads to **winner-take-most** outcomes
 - Hard for new competitors to match a large established network
 - **Examples**
 - Facebook beating new social networks
 - Uber dominating rides
 - Amazon dominating e-commerce
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WHEN NETWORK EFFECTS BECOME A MONOPOLY

- Once a platform is large enough, it becomes nearly impossible to displace.
 - **Examples**
 - Google Search (data + advertisers + users)
 - Windows in the 2000s (software availability)
 - VISA/Mastercard (merchant acceptance)
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QUICK COMPARISON

Type	Example	Impact
Direct	WhatsApp	High value with more users
Indirect	Daraz	Buyers ↔ Sellers boost each other
Data	TikTok	Better recommendations
Standard	QR	Universal acceptance

SUMMARY

- Network effects = foundation of digital platforms
- Direct, indirect, data, and technology types
- Critical mass → positive feedback loops
- Case studies in Nepal + global
- **Discussion question:**
 “What new platform in Nepal could successfully build strong network effects?”